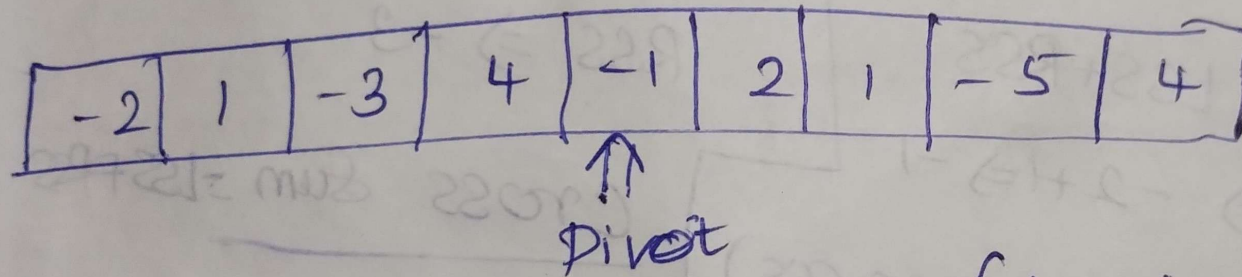


Maximum Subarray using Divide and Conquer

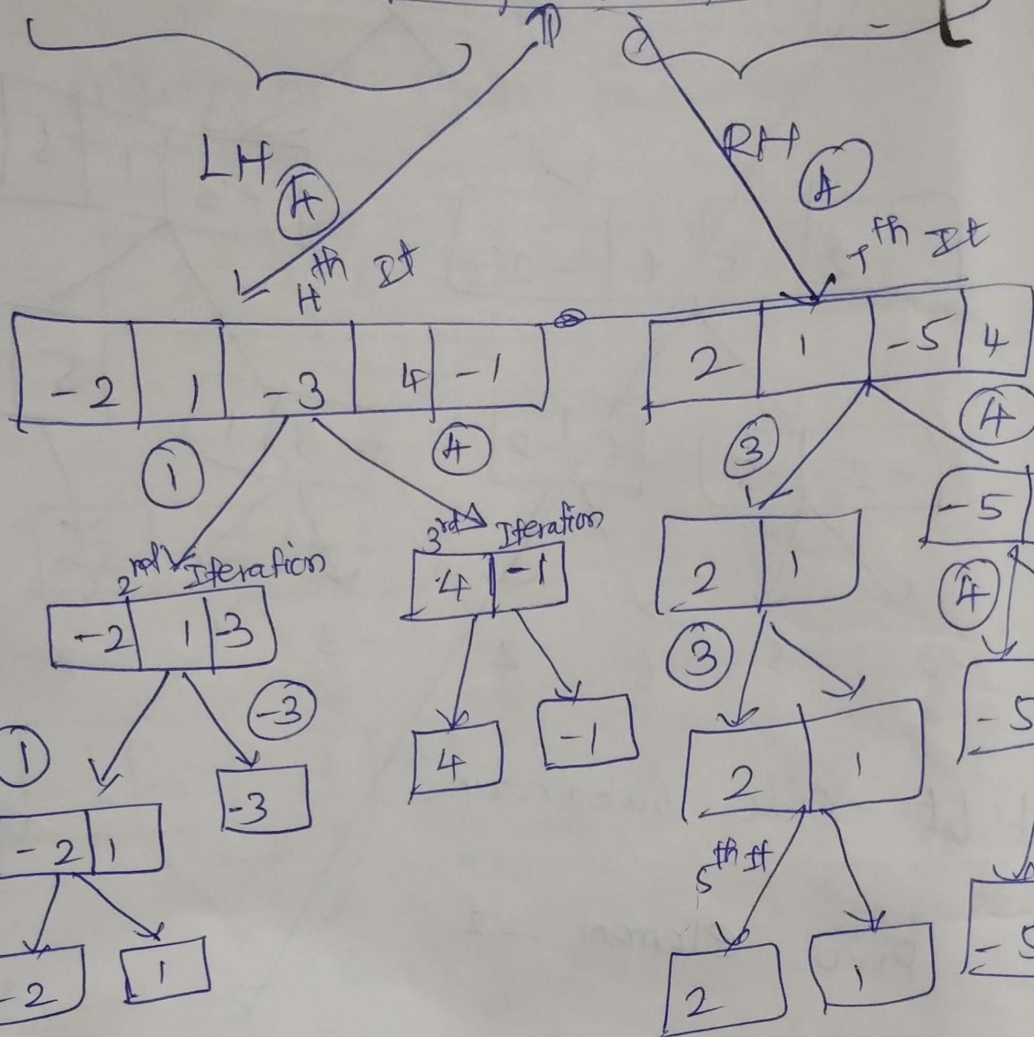
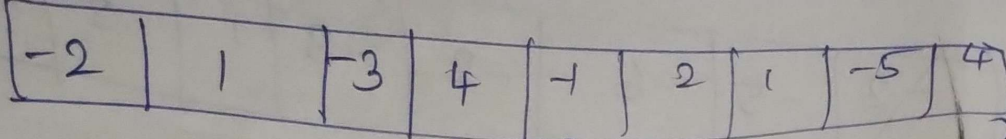
Input : -2, 1, -3, 4, -1, 2, 1, -5, 4



Index of Pivot $\Rightarrow \frac{(\text{begin} + \text{end})}{2} \Rightarrow \frac{0+8}{2} \Rightarrow 4$

Left half $\Rightarrow$ begin to pivot

Right half $\Rightarrow$ pivot + 1 to end



1st Iteration

Pivot = -2

Left Sum $\Rightarrow$ -2

Right Sum $\Rightarrow$ 1

Cross Sum $\Rightarrow$ LSS + RSS
 $\Rightarrow$ -2 + 1 $\Rightarrow$ -1

Max Sum $\Rightarrow$ Max(LSS, RSS, CSS)
 $\Rightarrow$ Max(-2, 1, -1)
 $\Rightarrow$ 1

2nd Iteration

Pivot = 1

LSS $\Rightarrow$ Max(1, (-2+1))

LSS $\Rightarrow$ 1

RSS $\Rightarrow$ -3

Cross Sum = LSS + RSS

$\Rightarrow$ 1 + (-3)

$\Rightarrow$ -2

Cross Sum = -2

Max Sum $\Rightarrow$ Max(LSS,

RSS, CSS) -
 $\Rightarrow$ Max(1, -3, -2) $\Rightarrow$ 1

3rd Iteration

$$LSS \Rightarrow 4$$

$$RSS \Rightarrow -1$$

$$CSS \Rightarrow LSS + RSS$$

$$\Rightarrow 4 - 1$$

$$\Rightarrow 3$$

$$\text{Max Sum}(LSS, RSS, CSS)$$

$$\Rightarrow \text{Max}(4, -1, 3)$$

$$\Rightarrow 4$$

$$\text{Pivot} = -3$$

$$LSS \Rightarrow \text{max}$$

$$4^{\text{th}} \text{ Iteration, } CSS \Rightarrow LSS + RSS$$

$$\Rightarrow -2 + 4 \Rightarrow 2$$

$$\text{Max}(2, 2, 4) \Rightarrow -2$$

$$\Rightarrow -2$$

$$RSS \Rightarrow \text{max}(4, 4-1)$$

$$\Rightarrow 4$$

$$\Rightarrow 4 \text{ is the left Sum}$$

5th Iteration

$$\text{Pivot} \Rightarrow 2$$

$$\text{Left Sum} = 2$$

$$\text{Right Sum} \Rightarrow 1$$

$$CSS \Rightarrow LSS + RSS$$

$$\Rightarrow 2 + 1$$

$$\Rightarrow 3$$

$$\text{Max}(LSS, RSS, CSS)$$

$$\Rightarrow (2, 1, 3)$$

$$\Rightarrow 3$$

6th Iteration

$$\text{Pivot} = -5$$

$$\text{Left Sum} \Rightarrow -5$$

$$\text{Right Sum} \Rightarrow 4$$

$$CSS \Rightarrow LSS + RSS$$

$$\Rightarrow -5 + 4$$

$$\Rightarrow -1$$

$$\text{Max}(LSS, RSS, CSS)$$

$$\Rightarrow \text{Max}(-5, 4, -1)$$

$$\Rightarrow 4$$

~~7th Iteration~~

$$CSS \Rightarrow LSS + RSS$$
$$\Rightarrow \text{Max}(3, 4)$$

$$\Rightarrow 4$$

$$RSS \Rightarrow 4$$

8th Iteration

$$\text{Left Sub Sum}$$

$$\text{Pivot} \Rightarrow -1$$

$$LSS \Rightarrow \text{Max}(-1, -1+4, -1+4-3)$$

$$-1, -1+4-3+1, -1+4-3+1-2$$

$$\Rightarrow \text{Max}(-1, 3, 0, 1, 1)$$

$$LSS \Rightarrow 3$$

RSS

$$\Rightarrow \text{Max}(2, 2+1, 2+1-5, 2+1-5+4)$$

$$\Rightarrow \text{Max}(2, 3, -2, 2)$$

$$\text{RSS} \Rightarrow 3$$

$$\text{CSS} \Rightarrow \text{LSS} + \text{RSS}$$

$$\text{CSS} \Rightarrow 6$$

Maximum Subarray Sum $\Rightarrow \text{Max}(\text{LSS}, \text{RSS}, \text{CSS})$

$$\Rightarrow \text{Max}(3, 3, 6)$$

$$\Rightarrow 6$$

7th Iteration

$$\text{LSS} \Rightarrow \text{Max}(-5, -5+1, -5+2)$$

$$\Rightarrow (-3)$$

$$\text{RSS} \Rightarrow 4$$

$$\text{CSS} \Rightarrow \text{LSS} + \text{RSS}$$

$$\Rightarrow 4-3$$

$$\Rightarrow 1$$

$$\text{Max}(\text{LSS}, \text{RSS}, \text{CSS})$$

$$\Rightarrow (-3, 4, 1) \Rightarrow 4$$